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Portfolio Optimization: Asset Universe Selection Based on Investor Profiles

Bachelor Thesis

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor Degree
- (ii) due acknowledgement has been made in the text to all other material used

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30 March, 2026

Acknowledgments

Acknowledgment goes here.

Abstract

Abstract text goes here.

Contents

Acknowledgments	III
Abstract	IV
1 Introduction	1
1.1 Motivation and Objectives	1
1.2 Outline	2
2 Literature Review	3
2.1 Framing the Problem	3
2.2 Why Pre-Selecting the Asset Universe Matters	3
2.3 How Asset Preselection Is Done in Practice	4
2.4 Investor Risk Tolerance and Personalized Investment Recommendation .	5
2.5 AI and Adaptive Methods	5
2.6 Synthesis and Research Gap	6
3 Methodology	7
3.1 Section Title 1	7
3.2 Section Title 2	7
4 Results & Limitations	8
4.1 Experiments Setup	8
4.2 Dataset Description	8
4.3 Results	8
4.3.1 Experiment 1	8
4.3.2 Experiment 2	8
4.4 Results Analysis and Discussion	8
5 Conclusion & Future Work	9
5.1 Conclusion	9
5.2 Future Work	9
Appendix	10

A Lists	11
List of Abbreviations	11
List of Figures	12
References	14

Chapter 1

Introduction

1.1 Motivation and Objectives

Portfolio construction is commonly framed as an optimization problem over a predefined set of assets. While this formulation is analytically convenient, it implicitly assumes that the same investable universe is suitable for all investors and remains appropriate across changing market conditions. In practice, this assumption is restrictive. Before portfolio weights can be meaningfully assigned, the candidate asset universe itself must first be evaluated for suitability with respect to investor needs and the current market environment.

This issue is particularly relevant in the Egyptian financial market, where asset risk behavior can shift materially over time in response to inflation dynamics, currency movements, monetary-policy changes, and episodes of market stress. Under such conditions, static categorizations that treat certain assets as permanently conservative or permanently aggressive may fail to reflect actual market behavior. As a result, portfolio construction that begins directly with final allocation may inherit weaknesses from an inappropriate or insufficiently adaptive candidate universe.

This thesis therefore treats asset-universe selection as a distinct decision stage within a broader investment pipeline. The focus is not the direct computation of final portfolio weights, but the prior problem of determining which assets should enter the investable set for a given investor context. The proposed direction is an artificial-intelligence-based framework, centered on reinforcement learning, that evaluates relative asset risk in a market-aware manner and supports adaptive universe construction instead of relying solely on fixed heuristics.

The main objective of the thesis is to design a scientifically grounded framework for AI-supported asset-universe selection in the Egyptian market. More specifically, the thesis aims to show that universe construction can be modeled as an explicit stage positioned between upstream risk quantification and downstream portfolio allocation. The defended contribution lies in combining AI-based selection, market-conditioned risk evaluation, and

the explicit treatment of asset-universe design as an independent component of portfolio construction rather than as a hidden preprocessing step. Accordingly, the central research question addressed by this thesis is: how can machine learning techniques support the dynamic selection of an investor-appropriate asset universe in the Egyptian financial market, based on risk tolerance, while allowing asset risk characteristics to adapt to changing market regimes?

1.2 Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on asset preselection, investor-risk-conditioned recommendation, adaptive AI methods, and the research gap addressed by this work. Chapter 3 presents the methodology and overall system design. Chapter 4 reports the results and evaluation of the proposed framework. Chapter 5 concludes the thesis and summarizes the main implications of the study.

Chapter 2

Literature Review

2.1 Framing the Problem

This thesis studies asset-universe selection as a stage that comes before final portfolio weighting. In practical terms, an optimizer can only allocate among the assets that are first admitted into the candidate set. If that set already contains assets that are too risky for the investor or poorly matched to current market conditions, then the later allocation step begins from a weak starting point. For this reason, the literature reviewed in this chapter is not limited to classical portfolio optimization papers. It also includes work on screening, personalization, and adaptive AI methods that help determine what should be considered investable in the first place.

Investor risk tolerance is one of the central ideas of this thesis, so the review gives special attention to papers in which recommendation outcomes change when investor characteristics change. This matters because the same market opportunity should not automatically be treated as equally suitable for every investor. At the same time, the review keeps a broad scope. Some papers recommend full portfolios, some recommend investment products or funds, and some learn dynamic financial policies directly. Even when those outputs are not identical to the output targeted here, they still help clarify how asset selection, investor profile, and adaptive AI can be connected in one decision process.

2.2 Why Pre-Selecting the Asset Universe Matters

A classical starting point for this discussion is the mean-variance framework introduced by Markowitz [1]. The framework explains how return and risk can be balanced when assigning portfolio weights, but it also relies on an assumption that is important for this thesis: the investable set is already known. In other words, the model answers how capital should be distributed across available assets, but it does not explain which assets

should first be considered suitable. This is useful here because it shows that asset-universe selection can be studied as its own decision stage instead of being treated as an invisible preprocessing step.

Later work makes that separation much more explicit. A deep-learning framework that preselects stocks before downstream portfolio formation was proposed by Wang et al. [2], while a CNN-BiLSTM approach that forecasts stock prices, ranks stocks, and then passes only the chosen subset into a Markowitz optimizer was used by Chaweewanchon and Chaysiri [3]. These papers are useful for this thesis because they show that preselection is not only about reducing dimensionality. In both cases, the screening stage is treated as a financially meaningful decision that can improve the quality of the later allocation stage.

The same idea appears in other forms as well. Machine-learning forecasts were used by Kaczmarek and Perez [4] to compare how filtered inputs behave under portfolio rules such as mean-variance optimization and hierarchical risk parity. Prediction models were also connected to portfolio design by Ma et al. [5], who showed that the final optimizer depends heavily on the quality of the assets that reach it. Another perspective was added through a two-stage robust framework that screens assets under uncertainty before solving the final construction problem, as shown by Mills and Anyomi [6]. Taken together, these papers support the same conclusion: a smaller but better-qualified asset set can make the later portfolio stage more meaningful and more robust.

2.3 How Asset Preselection Is Done in Practice

Although these papers share a staged structure, they do not all perform preselection in the same way. A predict-then-select pattern is followed in the studies of Wang et al. [2], Chaweewanchon and Chaysiri [3], Kaczmarek and Perez [4], and Ma et al. [5]. In these studies, the model first estimates future return, price movement, or another performance signal for each asset, then ranks the assets and forwards the strongest candidates to a later portfolio rule. This is useful for this thesis because it shows how AI can act as a filter before the optimizer begins assigning weights.

Preselection does not have to depend only on expected return, as shown by Mills and Anyomi [6]. Their framework screens assets under uncertainty and emphasizes robustness before the final allocation stage is reached. This is an important variation because it shifts the focus from “which asset looks best” to “which asset is qualified to move forward.” That distinction is relevant to this thesis, since investor risk tolerance is closer to a suitability question than to a pure return-maximization question.

The main practical lesson from this group of papers is that preselection usually produces a candidate set rather than a final investment answer. The output is often a shortlist or filtered universe that is later handled by an optimizer or recommendation rule. That role is close to the role defended in this thesis. However, there is still a clear limitation in this stream: the filtering stage is usually based on market information alone and rarely changes according to investor-specific risk tolerance. This leads naturally to the personalization literature.

2.4 Investor Risk Tolerance and Personalized Investment Recommendation

Once the selection problem is separated from the weighting problem, the next question becomes whose preferences should shape that selection. An early answer was provided through a case-based finance advisory system in which investor-specific information affects the advice and diversification strategy that are recommended, as shown by Musto et al. [7]. The output is not a formal asset universe, but the paper is still useful because it shows that recommendation quality changes when the investor is modeled explicitly rather than treated as generic.

A move toward AI-based personalization can be seen in the use of an adaptive neuro-fuzzy inference system to customize investment types from investor demographics and feedback, as reported by Asemi et al. [8]. This direction was later extended by Asemi et al. [9] through a combination of ANFIS, clustering, expert feedback, and multimodal neural network pretraining to recommend suitable investment types from investor preferences, goals, and experience. These papers matter for this thesis because they show a structured way to connect investor characteristics to financial recommendation. At the same time, the output remains an investment-type recommendation rather than a dedicated asset-universe filter.

A more adaptive form of personalization was introduced through a system that updates a customer's utility from repeated interactions and then uses that evolving signal to recommend funds from a predefined pool, as shown by Wei and Liu [10]. A recent supporting example was also provided by Schneider and Yilmaz [11], who showed that stock recommendations produced under different stated risk appetites can change across profiles. Together, these papers show that investor risk tolerance and investor behavior can influence what is recommended. The main limitation is that most of this literature still outputs product types, fund lists, or full portfolios rather than a clear middle stage that selects an investor-appropriate asset universe before a separate allocation step.

2.5 AI and Adaptive Methods for Risk-Aware Asset Selection

The personalization literature shows that investor-specific information matters, but it does not fully explain how an adaptive system should update its decisions over time. An important bridge was provided by a robo-advising framework that learns investor risk preferences from observed portfolio choices, as presented by Alsabab et al. [12]. This is useful for this thesis because it suggests that risk preference does not always need to be fixed in advance. A system can refine its understanding of the investor through interaction, which is a valuable idea for future extensions of investor-aware asset selection.

Adaptive financial AI is also widely studied through reinforcement learning. A foundational framework in which portfolio decisions are learned directly from market data

using shared evaluators, portfolio-vector memory, and an explicit reward structure was provided by Jiang et al. [13]. This direction was later made more accessible through FinRL, which offers environments, agents, and backtesting tools for financial reinforcement learning, as shown by Liu et al. [14]. These papers are not direct asset-selection papers, but they are useful because they show how dynamic financial decision systems are commonly designed and evaluated.

More recent work moves closer to risk-aware decision-making. Multiple PPO-based agents with different reward functions, including return, differential Sharpe ratio, and maximum drawdown, were trained and combined into a risk-adjusted policy by Choudhary et al. [15]. The closest overlap with the present thesis was then provided by Orra et al. [16], whose pipeline combines investor-specific risk categories, volatility-guided asset grouping, and deep reinforcement learning. These papers are useful because they show that adaptive AI can incorporate several risk-related objectives and investor-specific signals at the same time. However, the final target usually remains portfolio allocation, and in the case of Orra et al. [16], the work is still a preprint and the selection stage remains tightly linked to downstream construction rather than defended as an independent stage.

Overall, this supporting literature shows that adaptive AI methods can respond to both market variation and investor-specific information. However, most of the work still moves quickly from model input to final portfolio or trading decisions. The narrower problem of investor-aware asset-universe selection remains much less developed.

2.6 Synthesis and Research Gap

The reviewed literature leads to three main observations. First, classical and preselection papers show that the quality of the later portfolio stage depends on the quality of the assets that enter it. Second, personalization papers show that investor characteristics, especially risk tolerance, can change what should be recommended. Third, adaptive AI papers show that dynamic systems can respond to market information and sometimes to investor behavior as well. These three observations are individually well supported in the literature, but they are rarely combined in one clear framework.

The gap, therefore, is not simply that AI has not been used in finance. The narrower gap is that the literature only partially overlaps on the three ideas this thesis needs at the same time: preselection, investor-risk-conditioned recommendation, and adaptive AI. Preselection papers usually ignore investor profile. Personalized recommendation papers usually output investment advice, fund lists, or full portfolios rather than a dedicated asset universe. Adaptive reinforcement learning papers usually learn weights or trading actions directly. This thesis is positioned at that intersection. It studies how AI can support asset-universe selection as its own decision stage while keeping investor risk tolerance as a central input.

Chapter 3

Methodology

Introduction to chapter text.

3.1 Section Title 1

Section text.

3.2 Section Title 2

Section text.

Chapter 4

Results & Limitations

4.1 Experiments Setup

Section text.

4.2 Dataset Description

Section text.

4.3 Results

Section text.

4.3.1 Experiment 1

Subsection text.

4.3.2 Experiment 2

Subsection text.

4.4 Results Analysis and Discussion

Section text.

Chapter 5

Conclusion & Future Work

5.1 Conclusion

Section text.

5.2 Future Work

Section text.

Appendix

Appendix A

Lists

List of Figures

Bibliography

- [1] H. Markowitz, “Portfolio selection,” *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
- [2] W. Wang, W. Li, N. Zhang, and K. Liu, “Portfolio formation with preselection using deep learning from long-term financial data,” *Expert Systems with Applications*, vol. 143, p. 113042, 2020.
- [3] A. Chaweewanchon and R. Chaysiri, “Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning,” *International Journal of Financial Studies*, vol. 10, no. 3, p. 64, 2022.
- [4] T. Kaczmarek and K. Perez, “Building portfolios based on machine learning predictions,” *Economic Research - Ekonomska Istrazivanja*, vol. 35, no. 1, pp. 19–37, 2022.
- [5] Y. Ma, R. Han, and W. Wang, “Portfolio optimization with return prediction using deep learning and machine learning,” *Expert Systems with Applications*, vol. 165, p. 113973, 2021.
- [6] E. F. E. A. Mills and S. K. Anyomi, “A hybrid two-stage robustness approach to portfolio construction under uncertainty,” *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 9, pp. 7735–7750, 2022.
- [7] C. Musto, G. Semeraro, P. Lops, M. de Gemmis, and G. Lekkas, “Personalized finance advisory through case-based recommender systems and diversification strategies,” *Decision Support Systems*, vol. 77, pp. 100–111, 2015.
- [8] A. Asemi, A. Asemi, and A. Ko, “Adaptive neuro-fuzzy inference system for customizing investment type based on the potential investors’ demographics and feedback,” *Journal of Big Data*, vol. 10, p. 87, 2023.
- [9] A. Asemi, A. Asemi, and A. Ko, “A model for investment type recommender system based on the potential investors based on investors and experts feedback using ANFIS and MNN,” *Journal of Big Data*, vol. 11, no. 1, p. 128, 2024.
- [10] J. Wei and J. Liu, “Personalized fund recommendation with dynamic utility learning,” *Financial Innovation*, vol. 11, p. 48, 2025.

- [11] C. J. Schneider and Y. Yilmaz, “Stock portfolio selection based on risk appetite: Evidence from chatgpt,” *Finance Research Letters*, vol. 82, p. 107517, 2025.
- [12] H. Alsabah, A. Capponi, O. Ruiz Lacedelli, and M. Stern, “Robo-advising: Learning investors’ risk preferences via portfolio choices,” *Journal of Financial Econometrics*, vol. 19, no. 2, pp. 369–392, 2021.
- [13] Z. Jiang, D. Xu, and J. Liang, “A deep reinforcement learning framework for the financial portfolio management problem,” *arXiv preprint arXiv:1706.10059*, 2017.
- [14] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, and C. D. Wang, “Finrl: A deep reinforcement learning library for automated stock trading in quantitative finance,” *arXiv preprint arXiv:2011.09607*, 2020.
- [15] H. Choudhary, A. Orra, K. Sahoo, and M. Thakur, “Risk-adjusted deep reinforcement learning for portfolio optimization: A multi-reward approach,” *International Journal of Computational Intelligence Systems*, vol. 18, p. 126, 2025.
- [16] A. Orra, A. Bhambu, H. Choudhary, M. Thakur, and S. Natarajan, “Deep reinforcement learning for investor-specific portfolio optimization: A volatility-guided asset selection approach,” *arXiv preprint arXiv:2505.03760*, 2025.